

ICT703 – Module 8 Workshop

Learning Objectives

In this Workshop you will:

1. *Practice reading and writing a file*

Activity 1: Processing CSV Files

For this workshop, we are going to read and analyse some CSV files. Download the “Google Play Store.csv” data file from Canvas and save it into your folder. Note that for this workshop, ensure your data files are in the same folder as your program. Also make sure you have your working folder loading into Visual Studio Code (to ensure the data file is found).

Use the `csv.reader` method for the following:

- a) Read the file in and process all the lines. Print out the total number of lines of data in the file.
- b) Extend the above to determine how many apps were updated in 2018. HINT you can use string methods to find if it contains 2018.

Use the `csv.DictReader` method for the following:

- c) Read the file and print a list of unique categories. Use the constant ‘Category’ in your file reading.
- d) Extend the above to print a list of the unique categories and the count of rows that have that category.
- e) Extend the above by making it a function that takes a file name and a header to be more general.

Activity 2: Processing a Config File

Create a list of ICT703 students. A student has information such as name, degree and what was the last movie they watched.

Using a list of a dictionary for the above data structure

- a) Create a function that writes out this data structure to a file.
- b) Extend the program by creating a function that reads in the student list from a file.
- c) Delete the fixed data structure within your program and then see if you can read it in from a file, get an update for a student and then write out the new details.